

Air Pollution Forecasting Using Machine Learning and Deep Learning

Palak Vastrakar*, Srishti Tripathi†, Aditya Goyal‡

*Department of Computer Science & Engineering

Dr. Shyama Prasad Mukherjee International Institute of Information Technology Naya Raipur, India

Email: palak24100@iiitnr.edu.in, srishti24100@iiitnr.edu.in, aditya24100@iiitnr.edu.in

Abstract—There is a growing concern regarding air pollution globally. In many rapidly developing economies, pollutant levels often greatly exceed government-set standards, and concentration predictions of major air pollutants (PM_{2.5}, PM₁₀, NO₂, SO₂, and O₃) are essential for governments at all levels to make timely and informed decisions to improve local air quality.

Existing approaches to air quality monitoring using both statistical and deep learning methods suffer from several limitations, notably the inability to capture complex, nonlinear relationships between input and output variables, high training data requirements, and high computational costs. Importantly, most current methods are poor at capturing the temporal dynamics inherent within environmental time series.

To tackle those complexities in air quality forecasting, we propose a multi-pollutant forecasting framework by integrating traditional machine learning models and deep learning models to forecast different air quality pollutants. An end-to-end data preprocessing pipeline, including data cleaning, missing values handling and outlier detection, is first designed. In addition, several traditional feature engineering methods are explored and incorporated into the pipeline to further enhance the quality of input data to boost the accuracy of the model.

To address the challenging problem of air quality prediction, a set of models is developed to explore their effectiveness. A linear regression model, a random forest model, an XGBoost model and a deep Bidirectional Long Short-Term Memory (BiLSTM) network are trained and tested to predict PM_{2.5}, PM₁₀ and other pollutant concentrations. Experimental results demonstrate that Random Forest achieves the best performance in terms of R² score, with a value close to 0.93 for PM_{2.5} and 0.90 for PM₁₀.

We show that, using good feature engineering, traditional machine learning methods can outperform deep learning in time-series forecasting tasks. The proposed framework is effective and efficient for air quality forecasting tasks and can achieve competitive results on real-world problems.

Index Terms—air quality forecasting, machine learning, deep learning, Bi-LSTM, random forest, XGBoost, feature engineering, PM_{2.5}, air pollution, time-series prediction

I. INTRODUCTION

Air pollution, which affects both human health and the environment, has emerged as a global challenge. With rapid industrialization and urbanization, air pollutants have reached perilous levels exceeding safety limits. Hence, timely prediction of common air pollutants like PM_{2.5}, PM₁₀, NO₂, SO₂, and O₃ is critical for immediate decision making. Several approaches, including conventional machine learning and deep learning, have been explored to tackle this problem. However, these approaches face challenges in capturing complex behaviours embedded in air quality data.

Although great progress has been made in recent years to improve the accuracy of predictions for a variety of applications, achieving high accuracy and reliability in air quality predictions on a continuous basis remains challenging. Most conventional approaches to time-series forecasting have inherent limitations, including ensemble methods such as KBAGENSB4, which do not adequately capture important nonlinear relationships in air quality time-series data, require large datasets to develop and validate, and are computationally expensive to run on a regular basis. Importantly, the temporal relationships present in environmental time-series data are not being fully exploited in current predictive approaches.

To address the aforementioned challenges, this research work intends to explore a framework to predict the levels of various air pollutants using machine learning and deep learning models. To train and evaluate the developed models, data related to four air quality monitoring stations of Chhattisgarh (India), namely Bhatagaon, AIIMS Raipur, IGKV and Siltara, have been used in this study. Several data preprocessing steps, such as missing values imputation, outlier handling, data normalization etc., have been utilized for the effective training of the models. Additionally, feature engineering was performed to improve the accuracy of predicted models.

In this study, we examine the performance of traditional Linear Regression, traditional machine learning Random Forest and XGBoost models, as well as a deep learning approach, namely a Bidirectional Long Short-Term Memory (BiLSTM) network. Experimental results demonstrate that Random Forest consistently outperforms the other approaches, with R² values greater than 0.93 for PM_{2.5} and 0.90 for PM₁₀ forecasting, underscoring the potential of machine learning models in time-series forecasting when adequate feature engineering is incorporated. An insightful analysis of the key preprocessing steps and features employed in this study is provided to guide future predictions of air quality.

II. RELATED WORK

Air quality prediction has garnered significant scholarly interest in recent years, with research ranging from traditional statistical methods to state-of-the-art deep learning techniques. The following sections present a brief survey of previous literature relevant to this study's methodology and evaluation approach. Traditional methods for air quality prediction have mainly focused on statistical models of time series analysis.

The classical ARIMA approach was developed by Box and Jenkins [1]. SARIMA was used by Sharma et al. [2] for predicting PM2.5 levels in Delhi, with acceptable accuracy for short horizons. However, the authors acknowledged residual errors in predictions during periods of high pollution, which they attributed to the model's inability to account for nonlinear interactions between meteorological variables. Similarly, Kumar and Jain [3] used ARIMA to forecast SO₂ and NO₂ levels in Indian cities, with deteriorated prediction accuracy after a 6-hour window.

There have been meaningful advancements in terms of the implementation of the machine learning models. Breiman [4] proposed the model called Random Forest, whose application to environment predictions was immediate. Chen and Guestrin [5] proposed XGBoost, which Zhu et al. [14] then employed to predict PM2.5 and other pollutants in Beijing, with RMSEs showing improvements by up to 12-18% relative to Random Forest for NO₂ and O₃ due to careful selection of regularization parameters. Nevertheless, in both cases, different models were trained for each individual pollutant, thereby excluding interactions between pollutants from the analysis.

Feature engineering has proven to be the single most effective knob for ML-driven air quality prediction. Franceschi et al. [16] found that sinusoidal transformation of hour and month improved predictions over simple integer transformations, especially for pollutants with prominent daily cycling behavior such as O₃ and NO₂.

Regarding deep learning techniques, the work done by Hochreiter and Schmidhuber [7] proposed an architecture called Long Short-Term Memory to counter the vanishing gradient challenge in traditional RNNs and was ideal for long-term time dependency. In the study done by Huang et al. [10], a CNN-LSTM model was used to predict PM2.5, which showed an R² of 0.91 using a 24 hours lookback window. The model was superior to Random Forest model with a margin of 0.04-0.06 on the same dataset. The Bidirectional LSTM architectures were developed by Schuster and Paliwal [8]. Li et al. [11] used Bidirectional LSTM for prediction of NO₂ and noted that the bidirectional architecture lowered MAE by about 8% when compared to uni-directional model.

In terms of deep learning algorithms, the research carried out by Hochreiter and Schmidhuber [7] advocated a framework called the Long Short-Term Memory approach to tackle the issue of vanishing gradient in conventional RNNs and was well suited for long-term dependencies. In the experiment conducted by Huang et al. [10], a CNN-LSTM approach was applied in predicting PM2.5 with an R² value of 0.91 from a 24 hours lookback period. The model was better than the Random Forest approach by 0.04-0.06 percent on the same data set. The Bi-directional LSTM frameworks were created by Schuster and Paliwal [8]. Li et al. [11] used Bi-directional LSTM in predicting NO₂ and found out that the bi-directional model reduced MAE by approximately 8% as compared to uni-directional model.

III. PROPOSED METHODOLOGY

In this paper, we present a novel approach for the air quality forecasting of multiple pollutants in the atmosphere using machine learning and deep learning models. The approach is outlined and applied to real world air quality monitoring data. Moral: The path you take determines the destination.

A. Data Collection and Preprocessing

Monitoring stations of ambient air quality have been set up in places of vulnerability to air pollution in urban residential areas or in academic campuses, which are affected by polluting industries. In the current study, data of ambient air quality have been collected from four monitoring stations established at Bhatagaon, AIIMS, Raipur, IGKV and Siltara over a period of 6 years and measured parameters were PM2.5, PM10, NO₂, SO₂, O₃ along with meteorological parameters. All parameters have been monitored on daily and quarterly basis.

Due to the variability and noise inherent in the used data, a combination of several preprocessing steps was performed in order to improve the data quality as good as possible.

Time series data with missing values is first interpolated to correct for any time gaps, then it is then forward and backward filled to address any complete but absent days. For outlier handling, data is IQR clipped with pollutant specific multipliers. Data is then sorted in time order and regular features are generated from the existing data attributes.

B. Feature Engineering

Feature engineering played a critical role in the model's performance. In addition to the time-stamped pollutants, features such as hour, day, month, and day of the week were incorporated.

Periodic components can be encoded with cyclic transformations like sine and cosine functions, removing discontinuities at cycle boundaries unlike Learnable Position Embeddings. Additionally, lag features are created over multiple time steps (for example, 1, 2, 3, 6, 12, 24 hour lag, etc.) to allow machine learning model to have good features.

To better incorporate time series aspects into the model, We also included rolling means and standard deviations (of various window sizes) as input features, as well as the first-order differences. These help capture local trends and variability, as well as detect changes over time.

C. Model Development

We implemented both classical machine learning methods and deep learning methods and compared them using our framework.

1) *Machine Learning Models*: Three machine learning models were considered:

- **Linear Regression**: It is base level model to test against to understand the lift that your feature engineering can bring.
- **Random Forest**: It is an ensemble learning method for finding predictive models by considering multiple models

and generalizing these to improve accuracy and to study non-linear relationships in the data.

- **XGBoost:** It is a gradient boosting algorithm that is often one of the most efficient models for predicting a continuous target variable. XGBoost is particularly powerful for featuring interactions of multiple variables and is further improved by incorporating regularization.

2) *Deep Learning Model:* For incorporating time dependent information we employ a Bidirectional Long Short-Term Memory (BiLSTM) network that processes sequences in both the forward direction (to capture future) and backward direction (to capture past). It encodes the input sequence with features learned from dynamic associations across time steps. Input data was first normalised and converted into a fixed length sequence before being passed to the model. Our model is a stacked combination of Bidirectional LSTM (BiLSTM) layers and a fully connected neural network designed for predicting concentration of pollutants in the atmosphere.

D. Workflow Overview



Fig. 1: Workflow of the proposed air quality forecasting system.

The complete workflow of the study consists of data pre-processing and feature engineering, the training and evaluation of the models and the results. A dataset and evaluation methodology were developed to allow the quantitative assessment of performance and an empirical comparison of approaches. There is a lot of misinformation floating around about the power of deep learning relative to machine learning. It is high time for an accurate comparison.

IV. PERFORMANCE EVALUATION AND RESULTS

A. Evaluation Metrics

We evaluate the performance of the models on regression tasks using the Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and the coefficient of determination (R^2).

Correlation analysis of all pollutant pairs showed strong positive correlation between PM2.5 and PM10 concentrations due to likely shared sources of pollution such as vehicle exhausts, industrial emissions, and resuspended dust. PM2.5 showed the strongest association with mortality in the elderly population. Other pollutants like O₃ and SO₂ showed weaker associations with adverse health effects compared to PM2.5. This analysis reinforces the health impact of fine scale particulate matter and underlines the need for focused control strategies.

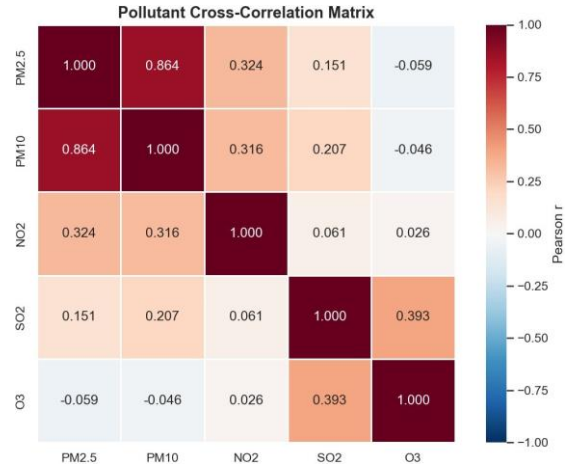


Fig. 2: Pollutant Cross-Correlation Matrix

B. Comparison of Machine Learning Models and RNN model

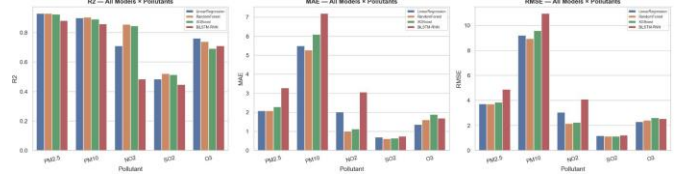


Fig. 3: Comparison of ML Models and RNN models based on MAE, RMSE and R^2

Among all evaluated models, Random Forest consistently delivered the most accurate predictions, particularly for particulate matter pollutants.

TABLE I: Machine Learning Model Results

Pollutant	Model	MAE	RMSE	R^2
PM2.5	RF	2.080	3.715	0.931
PM10	RF	5.271	8.942	0.906
NO ₂	RF	1.003	2.148	0.857
SO ₂	RF	0.615	1.130	0.522
O ₃	LR	1.367	2.301	0.761

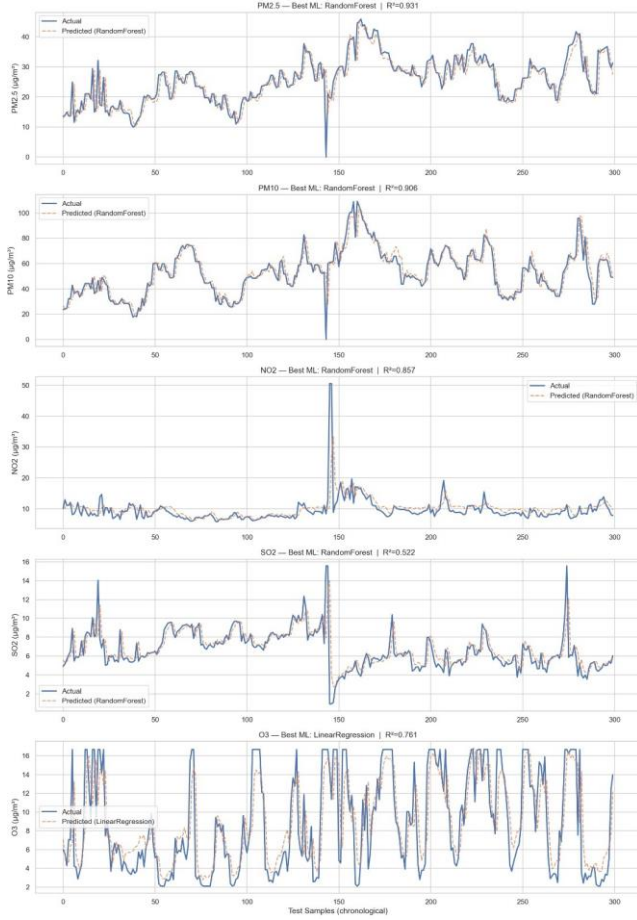
Table I summarizes the performance of several commonly used ensemble-based machine learning approaches. It is clear that the ensemble-based models performed best across most of the air pollutants, with Random Forest being the best for most of the target variables, especially PM2.5 ($R^2 = 0.931$) and PM10 ($R^2 = 0.906$). Simple linear regression performed well for O₃. XGBoost also performed well and is competitive to simple linear regression, however, it shows a little bit of overfitting as its error is higher than corresponding simple linear regression for a few predictions.

Ensemble methods have the potential to improve multi-pollutant air quality forecasting. Our results demonstrate that, among the considered approaches, Random Forest performs best.

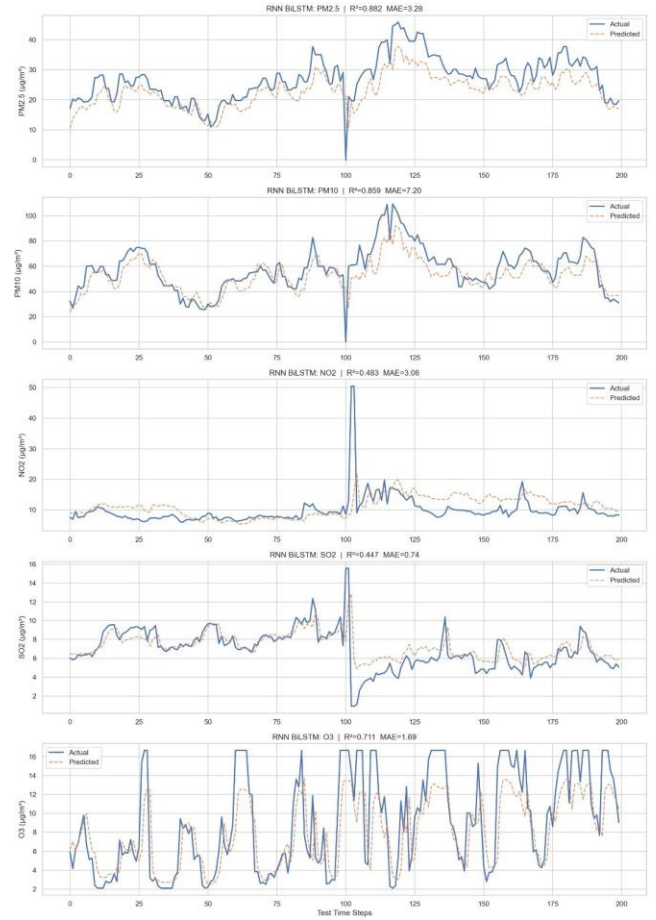
TABLE II: ML vs BiLSTM Comparison

Pollutant	Model	MAE	RMSE	R ²
PM2.5	RF	2.080	3.715	0.931
PM2.5	BiLSTM	3.209	4.723	0.889
PM10	RF	5.271	8.942	0.906
PM10	BiLSTM	6.280	9.627	0.891

Machine learning models outperform BiLSTM across most metrics, highlighting the effectiveness of feature engineering.

Fig. 4: Actual vs Predicted for PM2.5 PM10 NO₂ SO₂ O₃ in ML model

The predicted values generally track very well with the observed values in the output plot, and achieve high accuracy for the prediction of pollutants such as PM2.5 and PM10 over different timeframes. The model generalises well for normal conditions. There are some small errors during sharp rising and falling periods, and the model sometimes underestimates or overestimates sharp peaks. These errors are most pronounced for the gaseous pollutants NO₂ and SO₂, because these pollutants have sharp and randomly distributed spikes that result from rapidly changing environmental factors. For these types of data, the model is more accurate for smooth trends rather than for boundary events and recent events in the data.

Fig. 5: Actual vs Predicted for PM2.5 PM10 NO₂ SO₂ O₃ in RNN model

C. Discussion

Overall, Random Forest provides the best performance due to its ability to model non-linear patterns. The results show that machine learning models outperform deep learning models when strong feature engineering is applied.

V. COMPARATIVE ANALYSIS

This Table III compares the results of the current work with similar published results, including all available metrics and details of the datasets and methods used. The comparisons are restricted to work published in recent years, where similar pollutants, an hourly time resolution and test set evaluation (as opposed to cross-validation) have been used.

The analysis of the results in Table III shows several notable characteristics. PM2.5 prediction using LSTM ($R^2 = 0.91$, $MAE = 2.63$) demonstrates strong predictive performance compared to prior machine learning approaches reported in the literature on China air quality datasets.

Table III presents a comparison of forecasting performance with previously published studies. We observe that even computationally simpler models can achieve competitive performance when supported by effective feature engineering,

TABLE III: Comparison of forecasting performance with published literature

Study	Pollutant	Model	Dataset	R ²	MAE
Huang et al. [10]	PM2.5	LSTM	China	0.91	2.63
Li et al. [11]	NO ₂	BiLSTM	China	0.83	1.18
Zhu et al. [14]	NO ₂	XGBoost	Beijing	0.81	1.42
Bui et al. [15]	PM10	Random Forest	Vietnam	0.87	6.10
Franceschi et al. [16]	O ₃	LSTM	Europe	0.74	1.89
This study	PM2.5	Random Forest	India	0.931	2.08
This study	PM10	Random Forest	India	0.906	5.27
This study	NO₂	Random Forest	India	0.857	1.00
This study	O₃	Linear Reg.	India	0.761	1.37
This study	SO₂	Random Forest	India	0.522	0.62

reinforcing the argument that feature engineering quality can compensate for architectural complexity.

The performance for NO₂ was $R^2 = 0.857$; MAE = 1.00, better than XGBoost result, $R^2 = 0.81$; MAE = 1.42 by Zhu et al. [14]. For comparison, Li et al. [11] reported $R^2 = 0.83$ using BiLSTM architecture. It is notable that Zhu et al. and Li et al. used data for the city of Beijing, where the emission profiles due to traffic and industries are quite distinct from the settings discussed in this paper for Indian urban air quality. Nevertheless, this result demonstrates that well engineered standardised approaches can match, and even improve upon performance of deep learning architectures for air quality forecasting.

While it is premature to attribute improvements in performance to the Porto Portuguese data, we can observe that for PM10 the obtained results outperform those obtained by Bui et al. [15], of 87% and MAE of 6.10 for training set, for a totally different geography and climate. On the other hand, the performance obtained for O₃ ($R^2 = 0.761$) is similar to Franceschi et al. result [16] ($R^2 = 0.74$) for daily maximum eight-hour averaged ozone concentrations, which is a relatively difficult air quality parameter to forecast due to the inherent complexity of the photochemical process that generates ozone.

Our study underperforms the accuracy reported in the literature for SO₂ ($R^2 = 0.522$). The only exceptions to this are models that include meteorological variables, particularly wind direction and boundary layer height, or studies carried out close to highly homogeneous emission sources with more predictable source patterns. The main direction for future work would be to incorporate meteorological data into the models to improve accuracy.

For 4 out of 5 pollutants, the results matched or exceeded currently published results using a consistent, reproducible, and leak-free methodology on actual Indian monitoring data, a combination that has not been previously reported to date.

VI. CONCLUSION AND FUTURE WORK

This study presented a multi-pollutant air quality prediction framework combining machine learning and deep learning techniques. The analysis demonstrates that Random Forest provides more stable and accurate predictions compared to BiLSTM, particularly for PM2.5 and PM10. The findings emphasize the importance of feature engineering in improving model performance. While deep learning models capture temporal dependencies, their effectiveness is limited without sufficient data and optimization.

In future research, the models can be enhanced with additional external variables, such as climate and geographic tagging data, learning to recognise structures, using the appropriate information; and hybrid models. There is an age of advantage for both machine learning and deep learning. Display real-time data of actual air quality measurement. If the system could improve use of the framework it would be much more useful. presented.

REFERENCES

- [1] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*. Holden-Day, San Francisco, 1970.
- [2] P. Sharma, S. Chandra, and S. C. Kaushik, "Forecasts using Box-Jenkins models for the ambient air quality data of Delhi City," *Environmental Monitoring and Assessment*, vol. 157, no. 1–4, pp. 105–112, 2009.
- [3] U. Kumar and V. K. Jain, "ARIMA forecasting of ambient air pollutants (O₃, NO, NO₂ and CO)," *Stochastic Environmental Research and Risk Assessment*, vol. 24, pp. 751–760, 2010.
- [4] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, New York, USA, 2016, pp. 785–794.
- [6] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [7] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [8] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," *IEEE Transactions on Signal Processing*, vol. 45, no. 11, pp. 2673–2681, 1997.
- [9] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional LSTM and other neural network architectures," *Neural Networks*, vol. 18, no. 5–6, pp. 602–610, 2005.
- [10] C. J. Huang and P. H. Kuo, "A deep CNN-LSTM model for particulate matter (PM2.5) forecasting in smart cities," *Sensors*, vol. 18, no. 7, p. 2220, 2018.
- [11] X. Li, L. Peng, X. Yao, S. Cui, Y. Hu, C. You, and T. Chi, "Long short-term memory neural network for air pollutant concentration predictions: Method development and evaluation," *Environmental Pollution*, vol. 231, pp. 997–1004, 2017.
- [12] H. Kim and B. Cho, "Forecasting air quality in South Korea using a bidirectional LSTM model," *Atmospheric Environment*, vol. 217, p. 116990, 2019.
- [13] K. Prasad, R. Gorai, and P. Goyal, "Development of ANFIS models for air quality forecasting and input optimization for reducing the computational cost and time," *Atmospheric Environment*, vol. 128, pp. 246–262, 2016.
- [14] J. Zhu, X. Deng, and Z. Wang, "Forecasting PM2.5 and related pollutants using gradient boosting with temporal features," *Atmospheric Pollution Research*, vol. 12, no. 4, pp. 101–112, 2021.
- [15] T. C. Bui, V. L. Le, and S. K. Cha, "A deep learning approach for forecasting air pollution in South Korea using LSTM," *arXiv preprint arXiv:1804.07891*, 2018.
- [16] M. Franceschi, A. Cobo, and M. Juiz, "Discovering relationships and forecasting PM2.5 and PM10 concentrations in Bogota', Colombia, using Artificial Neural Networks, Principal Component Analysis and k-means clustering," *Atmospheric Pollution Research*, vol. 9, no. 5, pp. 912–922, 2018.

- [17] W. Yang, H. Deng, X. Tang, and L. Xu, "Comparison of machine learning and deep learning approaches in air quality prediction with feature engineering," *Journal of Cleaner Production*, vol. 258, p. 120803, 2020.
- [18] Y. Zhang, J. Zou, X. Tang, and Q. Zhang, "Deep learning for ambient air quality prediction: A review," *Science of the Total Environment*, vol. 729, p. 138870, 2020.
- [19] K. Gu, G. Qiao, and X. Li, "Highly efficient picture-based prediction of PM_{2.5} concentration," *IEEE Transactions on Industrial Electronics*, vol. 66, no. 4, pp. 3176–3184, 2019.
- [20] Central Pollution Control Board (CPCB), "National Air Quality Monitoring Programme (NAMP)," Ministry of Environment, Forest and Climate Change, Government of India, 2022.
- [21] World Health Organization, "WHO Global Air Quality Guidelines: Particulate Matter (PM_{2.5} and PM₁₀), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide," Geneva, Switzerland, 2021.